

RotateK — Imms-eval Lite Ablation

Token-only vs. ThinK / SparK / RotateK at matched KV budgets

Setup. Qwen2.5-VL-7B-Instruct on the `lmms-lab/LMMS-Eval-Lite` splits of TextVQA, InfoVQA and ChartQA (500 samples each), greedy decoding, one A100-80GB per cell. Metrics are exact match (TextVQA), ANLS (InfoVQA) and relaxed accuracy (ChartQA). Within each framework all rows share a KV budget: VisionZip $\rho_t=0.28$ vs. $\rho_t=0.45$, $\rho_c=0.25$ (both $0.28\times$); FastV $\rho_t=0.25$ vs. $\rho_t=0.40$, $\rho_c=0.25$ (both $0.25\times$). VisionZip’s token ratio is `dominant+contextual`, with `contextual=0.05` throughout. Reproduce with `scripts/paper/eval/lite_sweep.py`.

	TextVQA	InfoVQA	ChartQA	Mean
VisionZip Token _{0.28}	79.60 ± 1.70	60.78 ± 2.02	62.80 ± 2.16	67.73
+ ThinK	76.96 ± 1.77	65.89 ± 1.96	72.00 ± 2.01	71.62
+ SparK	82.72 ± 1.56	69.39 ± 1.94	72.40 ± 2.00	74.84
+ RotateK	81.42 ± 1.62	69.22 ± 1.94	72.00 ± 2.01	74.21
FastV Token _{0.25}	83.12 ± 1.55	54.81 ± 2.09	60.20 ± 2.19	66.04
+ ThinK	79.20 ± 1.70	63.28 ± 2.01	69.20 ± 2.07	70.56
+ SparK	85.58 ± 1.44	66.67 ± 1.99	70.00 ± 2.05	74.08
+ RotateK	85.20 ± 1.46	66.71 ± 1.97	70.80 ± 2.04	74.24

What reproduces. Spending the KV budget on Key channels rather than on tokens alone is clearly better: against token-only pruning at the same budget, the channel methods gain +3.9 to +8.2 mean points. RotateK improves over ThinK by +2.6 (VisionZip) and +3.7 (FastV) mean points, reproducing the paper’s central channel-pruning comparison.

What does not. RotateK and SparK are *within noise of each other* here (-0.63 on VisionZip, $+0.16$ on FastV), whereas the paper reports RotateK ahead. With 500 samples per benchmark the per-cell standard error is 1.5–2.1 points, so a 0.6-point mean gap is not resolvable. This ablation also covers three benchmarks, where the paper averages five (adding DocVQA and VizWiz), and uses lite rather than full validation splits. Separately, the paper’s case against SparK is primarily about decode latency — its per-token masks preclude the shared fused kernel — which this accuracy-only ablation does not measure.

Reading the numbers. Values are percentages with the benchmark’s own standard error; means are unweighted over the three benchmarks. Raw per-cell outputs, including the full metric dictionaries, are in the run logs.